

Homework Assignments
ASEN 6062: Celestial Mechanics
Spring 2026

HW#1: Problems 1 - 7, Due Friday, February 6

HW#2: Problems 8-14, Due Friday, February 27

8. For the following, use the condition that a central configuration yields stationary points of the function $\mathcal{U}^2 I_P$ relative to each position vector of the constituent particles.
- Show that a system of N equal masses placed at the vertices of a regular polygon (an N -gon) will always form a central configuration. Evaluate the constant σ for a system in such a central configuration.
 - Show that you can add a body of arbitrary mass at the center of the above N -gon and that it will remain a central configuration. Evaluate the new constant σ for this central configuration.
 - Show that a system of 4 equal masses placed at the vertices of a regular tetrahedron will always form a central configuration.
9. We have seen that any planar central configuration can form a homographic solution where its scaling parameter α and rotation rate Ω , follow the equations of motion:

$$\begin{aligned}\ddot{\alpha} - \alpha\Omega^2 &= \frac{2\sigma^*}{\alpha^2} \\ \alpha\dot{\Omega} + 2\dot{\alpha}\Omega &= 0\end{aligned}$$

where $\sigma^* < 0$. This turns out to be an integrable dynamical system (identical to the 2BP) which we solved in class. Carry out a detailed discussion of the solution to this dynamical system for the case when the total angular momentum $\beta = 0$, (i.e., $\alpha^2\Omega = 0$). Be sure to cover all the possible cases associated with a negative, positive and zero total energy.

10. For the 3-body problem Lagrange (equilateral triangle) central configuration, show that the center of mass of the system can lie at any point within and on the borders of the equilateral triangle, depending on the relative masses of the bodies. Assume $m_i \geq 0$ and $m_1 + m_2 + m_3 = 1$.
11. For the 3-body problem, find all solutions for the Euler (co-linear) central configurations when the masses of the three particles are equal, $m_1 = m_2 = m_3$.

12. Show that a mass distribution of constant density σ in the shape of a sphere of radius R yields a gravitational potential of the form:

$$\mathcal{U} = - \begin{cases} \frac{\mathcal{G}M}{|\mathbf{r}|} & r > R \\ \frac{\mathcal{G}M}{2R^3} [3R^2 - r^2] & r \leq R \end{cases}$$

for a test particle at a position $\mathbf{r}$ relative to the center of the sphere, and where $M = (4\pi/3)\sigma R^3$. Recall that a differential element of the gravitational potential has the form $d\mathcal{U} = -\mathcal{G}dM/|\mathbf{r} - \boldsymbol{\rho}|$ where $\boldsymbol{\rho}$ is the position of the differential mass element. Hint: use spherical coordinates to carry out the integration.

13. For the above force potential, show that integrating this again over its mass distribution, $\frac{1}{2} \int \mathcal{U}(\mathbf{r}) dm$, yields the self-potential

$$\mathcal{U}_{11} = -\frac{3\mathcal{G}M^2}{5R} \quad (1)$$

where the quantities are as defined in the previous problem. Hint: use spherical coordinates to carry out the integration.

14. Write out expressions for the Full 2-Body Problem free energy (total energy minus self-potentials), angular momentum, total moment of inertia for two spheres with equal density σ and radii R_1 and R_2 resting on each other. Write a program/spreadsheet to compute the necessary spin rate for them to “fission,” i.e., enter orbit about each other assuming they are initially resting on each other, and compute the free energy of the resulting system, allowing the radius ratio R_1/R_2 to vary between 0 and 1. Show your results as a plot of radius ratio vs. free energy at “fission.” At what radius ratio does the fission system go from positive free energy to negative free energy?

15. Consider the Full 3BP, with three spheres of equal radius R and equal density. There are two resting configurations of interest for this problem, with the three bodies in mutual contact (the Lagrange configuration) and with the bodies in a straight line (the Euler configuration). The stability of each of these configurations can be investigated by analyzing equilibrium with the single degree of freedom being the angle between the two bodies at the ends. When the angle equals 180 degrees the system takes on the Euler configuration, and when it takes on 60 degrees it takes on the Lagrange configuration.
- Using the definition of equilibrium (both constrained and unconstrained), show that each of these configurations can be a relative equilibrium.
 - Find the angular momentum at which the Lagrange Resting configuration ceases to exist.
 - Find the angular momentum at which the Euler Resting configuration becomes stable (it is unstable for small angular momentum).
 - Show that the Euler Resting configuration always becomes stable before the Lagrange Resting configuration ceases to exist.
16. The self-potential of an ellipsoid of revolution ($\alpha = \beta$) can be found in closed form as $\mathcal{U}_{ii} = -\frac{3}{10}\mathcal{G}M^2\mathcal{I}$, where M is the total mass, $\mathcal{G}$ is the gravitational constant and

$$\mathcal{I} = \begin{cases} \frac{2}{\sqrt{\alpha^2 - \gamma^2}} \arccos\left(\frac{\gamma}{\alpha}\right) & \alpha = \beta > \gamma \\ \frac{2}{\alpha} & \alpha = \beta = \gamma \\ \frac{1}{\sqrt{\gamma^2 - \alpha^2}} \ln \left[\frac{\gamma + \sqrt{\gamma^2 - \alpha^2}}{\gamma - \sqrt{\gamma^2 - \alpha^2}} \right] & \gamma > \alpha = \beta \end{cases}$$

In the following calculations assume that the system has been normalized such that $M = \mathcal{G} = \alpha\beta\gamma = 1$. Then the moments of inertia of the body are $\frac{2}{5}\alpha^2$ and $\frac{1}{5}(\alpha^2 + \gamma^2)$.

For the oblate ($\alpha > \gamma$) and the prolate ($\gamma > \alpha$) cases plot the amended potential value as a function of H^2 for several different values of α/γ , both greater and less than unity. Make sure that you always use the

maximum moment of inertia for I_H , which will change as a function of α/γ .

Solve for the value of H^2 at which the total amended potential equals zero as a function of the ratio α/γ . Identify the extreme values (i.e., maximum and minimum values) of H^2 at which the total energy of the rotating system is zero.

17. Consider a collection of N spheres of equal size and density. Assume that we choose constants such that the gravitational constant and their diameter are unity ($\mathcal{G} = d = 1$). Then the volume of each sphere is $V = \pi/6$ and their mass is $M = \sigma V$. For such a collection of bodies, they will always achieve their minimum potential energy when they are all in mutual contact with each other, i.e., $r_{ij} = 1$ for all $1 \leq i < j \leq N$. However, this is physically not possible for $N \geq 4$.

- Make a table giving the minimum total potential energy (self plus mutual) for this collection of spheres for $N = 1, 2, 3, 4$. Leave two additional columns for the lower bound and heuristic limit, defined below, and fill them in as well.
- Derive an expression for the lower bound on the total potential energy as a function of N by assuming the bodies are all in mutual contact (even though this is physically impossible for $N > 4$).
- Evaluate a few possible physically realistic minimum total potential energies for $N = 5$ by placing the additional mass at different points of the minimum energy 4 body configuration. Note, this is just a problem in geometry, as you need to find the distance from the “fifth” body to the other 4 bodies. Compare these physically realistic potential energies with the lower bound derived above.
- A “heuristic” minimum energy configuration for $N \gg 1$ is to assume that all N bodies are packed to form a sphere of total mass NM with an approximate diameter of $\sim N^{1/3}$. Compare this limit with all of your previous calculations for N from 1 to 5.
- In the limit as $N \rightarrow \infty$ compare the ratio of the heuristic minimum energy configuration with the lower bound potential energy.

18. Starting from the bary-centric restricted $N + 1$ -body problem:

$$\begin{aligned}\ddot{\mathbf{r}} &= -\frac{\partial U}{\partial \mathbf{r}} \\ U &= \sum_{i=1}^N \frac{\mu_i}{|\mathbf{r} - \mathbf{r}_i|} \\ \sum_{i=1}^N \mu_i \mathbf{r}_i &= 0\end{aligned}$$

where $\mu_i = \mathcal{G}M_i$, derive the circular, restricted 3-body problem in a rotating reference frame:

$$\begin{aligned}\ddot{x} - 2\omega\dot{y} &= -\frac{\partial V}{\partial x} \\ \ddot{y} + 2\omega\dot{x} &= -\frac{\partial V}{\partial y} \\ \ddot{z} &= -\frac{\partial V}{\partial z} \\ V &= \frac{1}{2}\omega^2(x^2 + y^2) + \frac{\mu_1}{|\mathbf{r} - \mathbf{r}_1|} + \frac{\mu_2}{|\mathbf{r} - \mathbf{r}_2|}\end{aligned}$$

where $\omega = \sqrt{(\mu_1 + \mu_2)/|\mathbf{r}_2 - \mathbf{r}_1|^3}$ is the rotation rate of the primaries about each other.

Choose length and time scales to reduce this to the standard, non-dimensional form of the equations:

$$\begin{aligned}\ddot{x} - 2\dot{y} &= -\frac{\partial V}{\partial x} \\ \ddot{y} + 2\dot{x} &= -\frac{\partial V}{\partial y} \\ \ddot{z} &= -\frac{\partial V}{\partial z} \\ V &= \frac{1}{2}(x^2 + y^2) + \frac{1 - \mu}{\sqrt{(x + \mu)^2 + y^2 + z^2}} + \frac{\mu}{\sqrt{(x - 1 + \mu)^2 + y^2 + z^2}}\end{aligned}$$

where $\mu = \mu_2/(\mu_1 + \mu_2)$

Write out the Jacobi integral for this problem.

19. **Slightly Modified** Starting from the body-centered restricted $N + 2$ -body problem:

$$\ddot{\mathbf{r}} = \frac{\partial U}{\partial \mathbf{r}}$$

$$U = \frac{\mu_0}{|\mathbf{r}|} + \sum_{i=1}^N \mu_i \left[\frac{1}{|\mathbf{r} - \mathbf{r}_i|} - \frac{\mathbf{r}_i \cdot \mathbf{r}}{|\mathbf{r}_i|^3} \right]$$

derive the restricted Hill 3-body problem in a rotating reference frame:

$$\begin{aligned} \ddot{x} - 2\omega\dot{y} &= \frac{\partial V}{\partial x} \\ \ddot{y} + 2\omega\dot{x} &= \frac{\partial V}{\partial y} \\ \ddot{z} &= \frac{\partial V}{\partial z} \\ V &= \frac{1}{2}\omega^2 (3x^2 - z^2) + \frac{\mu_0}{|\mathbf{r}|} \end{aligned}$$

where ω is the rotation rate of the primaries about each other. To derive this form the function $\mu_1 \left[\frac{1}{|\mathbf{r} - \mathbf{r}_1|} - \frac{\mathbf{r}_1 \cdot \mathbf{r}}{|\mathbf{r}_1|^3} \right]$ must be expanded to higher orders in $1/|\mathbf{r}_1|$ (**assuming that** $|\mathbf{r}_1| \gg |\mathbf{r}|$) and we must assume that $\mu_0 \ll \mu_1$.

Choose length and time scales to reduce this to the standard, non-dimensional form of the equations:

$$\begin{aligned} \ddot{x} - 2\dot{y} &= \frac{\partial V}{\partial x} \\ \ddot{y} + 2\dot{x} &= \frac{\partial V}{\partial y} \\ \ddot{z} &= \frac{\partial V}{\partial z} \\ V &= \frac{1}{2} (3x^2 - z^2) + \frac{1}{|\mathbf{r}|} \end{aligned}$$

Write out the Jacobi integral for this problem.

20. The gravitational force potential of a 2nd degree and order gravity field can be expressed in spherical coordinates as:

$$U = \frac{\mu}{r} + \frac{\mu}{r^3} \left[C_{20} \left(1 - \frac{3}{2} \cos^2 \delta \right) + 3C_{22} \cos^2 \delta \cos 2(\lambda - \omega t) \right]$$

where δ is the latitude, λ is the inertial longitude, ω is the rotation rate of the body, and t is the time.

Using Lagrange's equations, write out the equations of motion in a body-fixed frame using spherical coordinates of radius, latitude, and longitude.

Write out the Jacobi integral for this problem.

21. Choose a particular astrodynamics or celestial mechanics problem of interest (i.e., N -body problem, restricted 3-body problem, Hill problem, orbiter problem, or some other variation) and implement it in a numerical integration package/routine. Also, define the equations of motion for the state transition matrix and incorporate them into your model. From this you should have the capability to numerically integrate a trajectory and state transition matrix from some initial conditions to a final time.

As part of this exercise you should be able to verify that your simulation conserves all or any relevant integrals of motion, that your numerically integrated state transition matrix verifies some of its basic properties, and you should be able to display the results of your integrations via plots or other such visualizations.

22. Show that the characteristic equation of a symplectic matrix Φ of dimension $2n \times 2n$, defined by $|\lambda I_{2n \times 2n} - \Phi| = 0$, has the following general form:

$$\begin{aligned} 0 &= |\lambda I_{2n \times 2n} - \Phi| \\ &= \lambda^{2n} + \alpha_1 \lambda^{2n-1} + \alpha_2 \lambda^{2n-2} + \dots + \alpha_{n-1} \lambda^{n+1} + \alpha_n \lambda^n + \alpha_{n-1} \lambda^{n-1} + \dots \\ &\quad + \alpha_2 \lambda^2 + \alpha_1 \lambda^1 + 1 \end{aligned}$$

i.e., that it is symmetric in its coefficients.

23. Show that the characteristic equation of the dynamics matrix of an arbitrary Hamiltonian system of dimension $2n$, $A = JH_{xx}$, not necessarily taken at an equilibrium, and defined by $|\sigma I_{2n \times 2n} - A| = 0$, has the following general form:

$$\begin{aligned} |\sigma I_{2n \times 2n} - A| &= 0 \\ &= a_{2n} \sigma^{2n} + a_{2n-2} \sigma^{2n-2} + \dots + a_2 \sigma^2 + a_0 \end{aligned}$$

i.e., that it only has even coefficients.